

7th Grade Life Science — Vocabulary Reference

Georgia Standards of Excellence (S7L1 – S7L5) • Five Unit Vocabulary

Name: _____

Period: _____

Date: _____

Section 1 — Classifying Life's Diversity

S7L1

Biology: the scientific study of living things.

Organism: a single, complete living thing.

Cell: the smallest unit of life; all living things are made of one or more cells.

Characteristics of Life: traits shared by all living things: made of cells, use energy, grow and develop, reproduce, respond to the environment, and maintain homeostasis.

Homeostasis: keeping a stable internal environment even when the outside changes.

Stimulus: a change in the environment that causes a response.

Response: what an organism does when it reacts to a stimulus.

Classification: sorting living things into groups based on shared traits.

Taxonomy: the science of naming and classifying living things.

Aristotle: an early scientist who first sorted living things into plants and animals.

Carolus Linnaeus: the scientist who created the modern naming and classification system.

Kingdom: the largest, broadest group used to classify living things.

Six Kingdoms: Archaeobacteria, Eubacteria, Protists, Fungi, Plants, and Animals.

Archaeobacteria: single-celled organisms that live in extreme environments.

Eubacteria: common single-celled bacteria found almost everywhere.

Protist: a mostly single-celled organism that is not a plant, animal, or fungus (e.g., amoeba, algae).

Fungi: organisms that absorb nutrients from other things (e.g., mushrooms, mold, yeast).

Taxonomic Levels: the order of classification groups: Domain, Kingdom, Phylum, Class, Order, Family, Genus, Species.

Genus: a classification group of closely related species; the first part of a scientific name.

Species: a group of organisms that can reproduce together; the most specific classification level.

Binomial Nomenclature: the two-word naming system using genus and species (e.g., *Canis lupus*).

Dichotomous Key: a tool that identifies an organism through a series of paired either/or questions.

Section 2 — Cells, Tissues & Body Systems

S7L2

Organelle: a small structure inside a cell that does a specific job ("little organ").

Nucleus: the control center of the cell; holds DNA and directs cell activities.

Cytoplasm: the jelly-like fluid that fills the cell and holds organelles in place.

Cell Membrane: the outer border of the cell; controls what enters and leaves.

Cell Wall: a stiff outer layer in plant cells that provides shape and support.

Mitochondria: the organelle that releases energy from food; the cell's powerhouse.

Chloroplast: a plant-cell organelle that captures sunlight to make food (photosynthesis).

Lysosome: the organelle that breaks down waste and worn-out parts.

Vacuole: a storage organelle for water, food, and waste; large and central in plant cells.

Photosynthesis: the process plants use to turn sunlight, water, and carbon dioxide into food and oxygen.

Tissue: a group of similar cells working together to do the same job.

Organ: different tissues working together as a unit (e.g., the heart).

Organ System: a group of organs that work together on a shared task.

Levels of Organization: the order from simple to complex: cell → tissue → organ → organ system → organism.

Cardiovascular System: pumps blood to carry oxygen and nutrients; includes the heart and blood vessels.

Respiratory System: brings in oxygen and removes carbon dioxide; includes the lungs.

Digestive System: breaks down food into nutrients; includes the stomach and intestines.

Excretory System: removes liquid waste from the body; includes the kidneys.

Muscular System: moves the body and its parts; made of muscles.

Nervous System: sends signals to control and sense; includes the brain and nerves.

Immune System: fights germs and disease; includes white blood cells.

System Interaction: two or more body systems working together to carry out a life process.

Section 3 — Reproduction & Heredity

S7L3

Trait: a feature that can be observed in a living thing (e.g., eye color).

Heredity: the passing of traits from parents to their offspring.

DNA: the chemical code that carries the instructions for all of an organism's traits.

Gene: a short section of DNA that controls one trait.

Chromosome: a long, coiled strand of DNA; genes sit along it.

Allele: a different version of a gene; offspring inherit two, one from each parent.

Dominant: the stronger allele; shown by a capital letter and hides the recessive allele.

Recessive: the weaker allele; shown by a lowercase letter; only appears when both alleles are recessive.

Genotype: the gene combination an organism has (e.g., Tt).

Phenotype: the physical trait that can be seen (e.g., can roll tongue).

Homozygous: having two of the same allele (TT or tt).

Heterozygous: having two different alleles (Tt).

Punnett Square: a tool that predicts the probability of offspring traits.

Probability: the chance that a particular outcome will happen.

Asexual Reproduction: reproduction needing only one parent; offspring are identical clones.

Clone: an offspring with exactly the same DNA as its single parent.

Sexual Reproduction: reproduction needing two parents; offspring are a unique mix of both.

Genetic Variation: differences in traits among the offspring produced by sexual reproduction.

Mutation: a change in DNA that can create a new trait.

Selective Breeding: when humans choose which organisms reproduce to pass on desired traits.

Section 4 — Ecology & Ecosystems

S7L4

Ecology: the study of how living things interact with each other and their environment.

Ecosystem: all the living and non-living things in an area and how they interact.

Biotic Factor: a living part of an ecosystem (plants, animals, fungi, bacteria).

Abiotic Factor: a non-living part of an ecosystem (sunlight, water, air, soil, temperature).

Population: all the members of one species living in the same area.

Community: all the populations of different species living together in an area.

Producer: an organism that makes its own food using sunlight (e.g., plants).

Consumer: an organism that gets energy by eating other organisms.

Decomposer: an organism that breaks down dead matter and returns nutrients to the soil.

Herbivore: a consumer that eats only plants.

Carnivore: a consumer that eats only other animals.

Omnivore: a consumer that eats both plants and animals.

Food Chain: a single path showing how energy moves from one organism to the next.

Food Web: all the connected food chains in an ecosystem.

Energy Pyramid: a diagram showing how energy decreases at each higher feeding level.

10% Rule: only about 10% of energy passes from one level to the next; the rest is lost as heat.

Water Cycle: the continuous movement of water through evaporation, condensation, and precipitation.

Carbon Cycle: the movement of carbon between the air, living things, and the soil.

Limiting Factor: anything that keeps a population from growing too large (resources, disease, climate, human activity).

Biome: a large region with its own climate, plants, and animals.

Estuary: an aquatic area where a freshwater river meets the salty ocean; very rich in life.

Marine Ecosystem: a saltwater ecosystem such as the ocean or a coral reef.

Section 5 — Evolution & Natural Selection

S7L5

Evolution: the change in the inherited traits of a species over many generations.

Adaptation: an inherited trait that helps an organism survive in its environment.

Variation: the small differences in traits among members of the same species.

Natural Selection: the process where organisms with helpful traits survive and reproduce more often.

Charles Darwin: the scientist who first described natural selection (1859).

Fitness: how well an organism is suited to survive and reproduce in its environment.

Survival of the Fittest: the idea that the best-suited organisms are most likely to survive and reproduce.

Competition: the struggle between organisms for limited resources like food and space.

Inheritance: the passing of helpful traits from surviving parents to their offspring.

Selective Pressure: an environmental factor that influences which traits are favored.

Extinction: when an entire species dies out and no longer exists.

Fossil: the preserved remains or trace of a living thing from long ago.

Fossil Record: the full collection of fossils, used as a timeline of life on Earth.

Rock Layers (Strata): layers of rock where deeper layers, and their fossils, are older.

Relative Age: the age of a fossil compared to others, based on which rock layer it is found in.

Diversity: the variety of different species; it has increased over time.

Common Ancestor: an organism from which different modern species have descended.

Peppered Moth: a classic example of natural selection, where moth color shifted as trees darkened.